

RETAILPULSE–AI-POWERED CUSTOMERANALYTICS&DEMAND FORECASTING PLATFORM

PROJECT REPORT

Submitted in partial fulfillment of the requirements for the
Data Science & Analytics Internship Program

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CERTIFICATE

This is to certify that the project entitled **“RetailPulse – AI-Powered Customer Analytics & Demand Forecasting Platform”** has been successfully completed by **Ms. Khushi Chauhan** as a part of the Data Science & Analytics Internship Program conducted by Zidio Development during the academic year 2025–2026.

The work presented in this project report is an original record of the candidate’s work carried out during the internship period under the guidance and support provided by Zidio Development.

This project fulfills the requirements of the internship program and demonstrates practical implementation of **Data Analytics, Machine Learning, Sales Forecasting, Customer Segmentation, and Dashboard Development using Python and Streamlit technologies.**

To the best of our knowledge, this project has not been submitted previously for any degree, diploma, certification, or any other academic purpose.

Place: _____

Project Mentor

Date: _____

Zidio Development

DECLARATION

I hereby declare that the project report entitled “**RetailPulse – AI-Powered Customer Analytics & Demand Forecasting Platform**” submitted to Zidio Development as a part of the Data Science & Analytics Internship Program is an original work carried out by me during the academic year 2025–2026.

I further declare that this project has not been submitted previously, either fully or partially, for the award of any degree, diploma, certification, or any other academic purpose.

The information and results presented in this report are true to the best of my knowledge and belief.

Place: _____

Signature

Date: _____

Khushi Chauhan

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Zidio Development for providing me the opportunity to work on the **project “RetailPulse – AI-Powered Customer Analytics & Demand Forecasting Platform”** under the Data Science & Analytics Internship Program.

I would also like to thank all the mentors and supporters who guided me throughout the development of this project. Their valuable suggestions, encouragement, and support helped me successfully complete this work.

This project has enhanced my practical knowledge in **Python, Machine Learning, Data Analytics, Data Visualization, and Streamlit Dashboard Development.**

I am thankful to everyone who contributed directly or indirectly towards the successful completion of this project.

Khushi Chauhan

ABSTRACT

RetailPulse is an AI-powered customer analytics and demand forecasting platform developed to help retail businesses analyze sales performance, understand customer behavior, forecast future demand, and optimize inventory management.

The system is developed using **Python, Streamlit, and Machine Learning algorithms** such as **KMeans Clustering, Random Forest Classifier, and Prophet Forecasting Model**.

The platform provides an interactive dashboard for visualizing sales trends, customer segmentation, churn prediction, and inventory recommendations. By using data analytics and forecasting techniques, RetailPulse helps businesses make data-driven decisions and improve operational efficiency.

This project demonstrates the **practical implementation of machine learning, business analytics, and data visualization in the retail industry**.

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1. INTRODUCTION

In today's digital era, retail businesses generate a massive amount of sales and customer-related data daily. Managing and analyzing this data manually becomes difficult and time-consuming. Businesses require intelligent systems that can transform raw data into meaningful insights for improving operational efficiency and customer satisfaction.

RetailPulse is an AI-powered retail analytics and demand forecasting platform developed using **Python and Streamlit**. The system uses data analytics and machine learning techniques to analyze retail sales data, identify customer purchasing patterns, forecast future demand, and provide inventory recommendations.

The platform offers an interactive dashboard that enables retailers to monitor business performance through visual reports and predictive analytics. RetailPulse helps businesses make data-driven decisions, improve inventory management, enhance customer retention, and optimize sales strategies.

2. PROBLEM STATEMENT

Retail businesses often face several challenges such as:

- Difficulty in predicting future sales
- Poor inventory management
- Lack of customer behavior analysis
- Manual data analysis is consuming significant time
- In ability to identify high-value customers
- In efficient business decision-making

Retail Pulse solves these problems using machine learning and data analytics techniques

3. OBJECTIVES

The main objectives of the project are:

- To analyze retail sales data
- To visualize monthly sales trends
- To identify top-performing products
- To perform customer segmentation using machine learning
- To forecast future sales using Prophet model
- To predict customer churn
- To recommend inventory stock actions
- To create an interactive Streamlit dashboard

4. TECHNOLOGIES USED

Technology	Purpose
Python	Programming Language
Pandas	Data Processing
NumPy	Numerical Computation
Matplotlib	Data Visualization
Streamlit	Interactive Dashboard
Scikit-learn	Machine Learning
Prophet	Sales Forecasting
Joblib	Model Saving
GitHub	Version Control

5. DATASET DESCRIPTION

The dataset used in this project contains retail sales information such as:

- Order Date
- Product Name
- Category
- Sales
- Profit
- Quantity
- Customer Name
- Region

The dataset was cleaned and preprocessed before analysis.

Data Cleaning Steps

- Removed missing values
- Converted sales columns into numeric format
- Converted order dates into datetime format
- Removed invalid records
- Standardized column names

Data preprocessing improved the quality and accuracy of machine learning models and analytics results.

6. SYSTEM ARCHITECTURE

The workflow of RetailPulse is as follows:

1. Data Collection
2. Data Cleaning & Preprocessing
3. Exploratory Data Analysis
4. Machine Learning Model Training
5. Dashboard Visualization
6. Forecast Generation
7. Customer Segmentation
8. Inventory Recommendation

7. METHODOLOGY

7.1 Data Collection

Retail sales data was collected from a structured dataset containing customer transactions, product details, and sales information.

7.2 Data Preprocessing

The collected dataset was cleaned and transformed before analysis.

Steps Included

- Handling missing values
- Removing duplicate records
- Formatting date columns
- Data type conversion
- Standardizing column names

7.3 Exploratory Data Analysis (EDA)

EDA techniques were used to identify trends, patterns, and business insights through charts and graphs.

7.4 Machine Learning Model Development

Different machine learning algorithms were implemented for:

- Customer segmentation
- Sales forecasting
- Churn prediction

7.5 Dashboard Development

An interactive dashboard was developed using Streamlit for data visualization and business analytics.

8. FEATURES OF RETAILPULSE

8.1 Sales Overview

The dashboard displays:

- Total Sales
- Total Orders
- Total Profit

These metrics help businesses understand overall performance.

8.2 Monthly Sales Trend

Monthly sales trends are visualized using line charts to identify seasonal patterns and revenue fluctuations.

Benefits

- Detect high sales periods
- Understand business growth
- Support business planning

8.3 Top 10 Products Analysis

The dashboard identifies top-performing products based on sales quantity.

Benefits

- Identify best-selling products
- Improve product marketing
- Focus on profitable items

8.4 Category Sales Analysis

Sales are grouped by product categories to determine which category generates maximum revenue.

Benefits

- Better category management
- Improved business strategy

8.5 Customer Segmentation

Customer segmentation is performed using:

- RFM Analysis
- K Means Clustering

RFM Components

Component	Meaning
Recency	Last purchase timing
Frequency	Number of purchases
Monetary	Total spending

Customers are classified into:

- VIP Customers
- Regular Customers
- Low Value Customers

Benefits

- Personalized marketing
- Better customer retention
- Improved targeting strategy

8.6 Sales Forecasting

Future sales are predicted using Facebook Prophet forecasting model.

Forecasting Process

1. Prepare time-series data
2. Train Prophet model
3. Generate future predictions
4. Display forecast graph

Benefits

- Future demand estimation
- Better inventory planning
- Improved business forecasting

8.7 Churn Prediction

A Random Forest Classifier is used to predict customer churn.

Features Used

- Sales
- Quantity
- Discount

Output

The model predicts whether customers are likely to churn.

Benefits

- Customer retention
- Marketing optimization
- Risk analysis

8.8 Inventory Recommendation System

Inventory recommendations are generated based on average sales quantity.

Recommendations

- Increase Stock
- Maintain Stock

Benefits

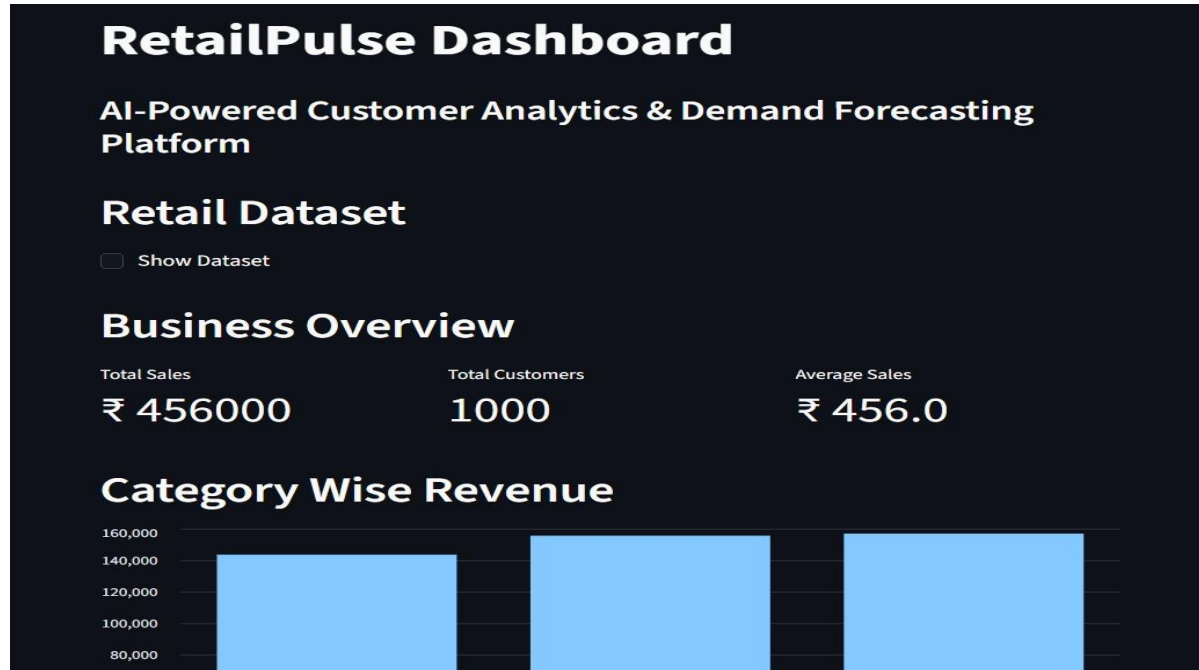
- Reduce stock shortages
- Better inventory planning
- Improve operational efficiency

9. MACHINE LEARNING MODELS USED

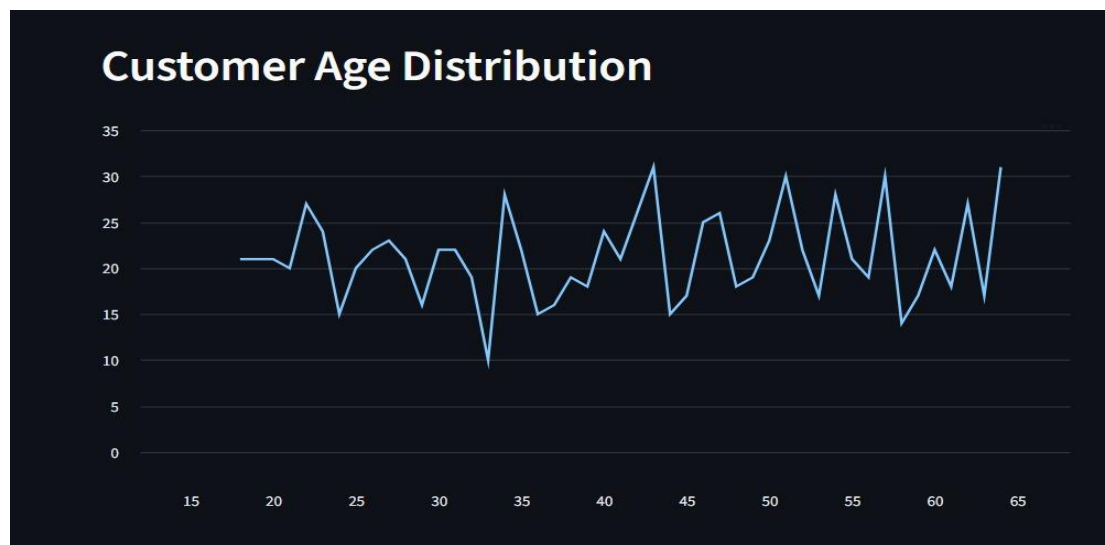
Model	Purpose
K Means Clustering	Customer Segmentation
Prophet	Sales Forecasting
Random Forest Classifier	Churn Prediction

10.DASHBOARD SCREENSHOTS

Home Dashboard



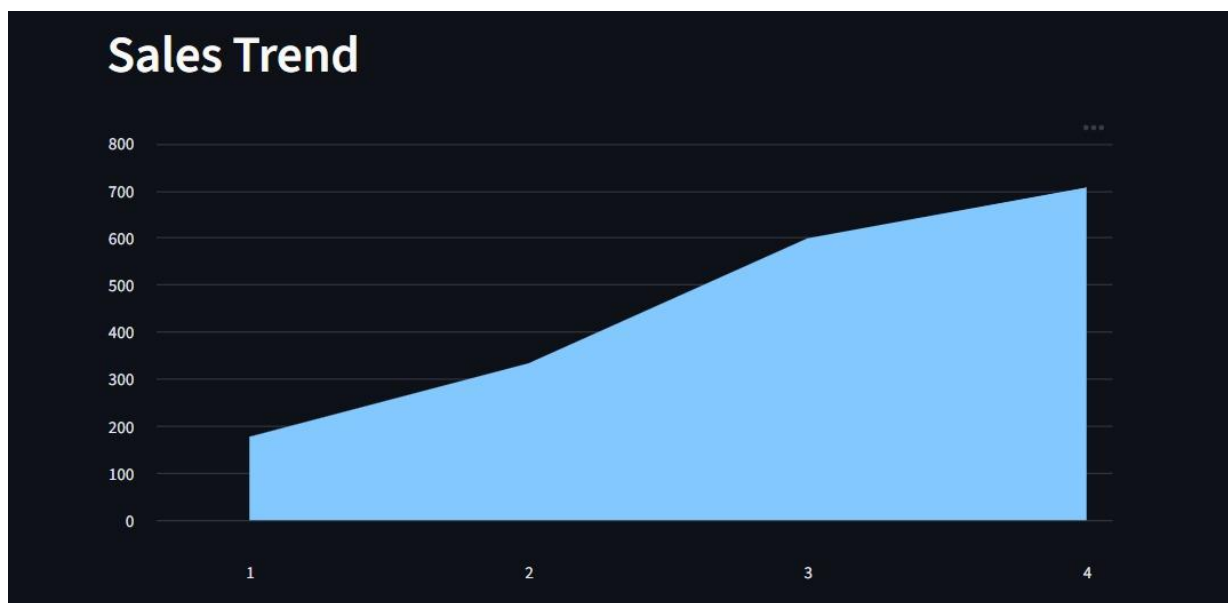
Customer Age Distribution



Category Wise Revenue



Sales Trend



Top 10 Customers



Predict Future Sales

CUST447	2000
CUST342	2000

Predict Future Sales

Enter Quantity

1 - +

Enter Price Per Unit

100 - +

Enter Customer Age

25 - +

Predict Sales

Customer Type Prediction

Check Customer Type

Customer Type Prediction

Customer ID	Total Amount
CUST015	2000
CUST412	2000
CUST970	2000
CUST547	2000
CUST281	2000
CUST416	2000
CUST420	2000
CUST927	2000
CUST447	2000
CUST342	2000

12. ADVANTAGES OF RETAILPULSE

- Reduces manual data analysis
- Improves business decision-making
- Enhances customer understanding
- Provides future sales predictions
- Helps optimize inventory levels
- Supports business growth strategies
- User-friendly interactive dashboard

13. CHALLENGES FACED

During development, the following challenges were encountered:

- Data cleaning and preprocessing issues
- Handling missing values
- Forecasting model configuration
- Streamlit integration errors
- Managing machine learning model files
- Visualization formatting issues

These challenges were resolved using proper debugging and preprocessing techniques.

14. FUTURE IMPROVEMENTS

Future enhancements for RetailPulse include:

- Real-time database integration
- Advanced deep learning forecasting models
- Cloud deployment using AWS
- Mobile-responsive dashboard
- Automated email reports
- Real-time inventory alerts
- User authentication system

15. CONCLUSION

RetailPulse successfully demonstrates the implementation of artificial intelligence and data analytics in the retail industry. The system integrates machine learning models, forecasting techniques, and interactive visualizations into a single dashboard platform.

The project provides valuable business insights such as sales analysis, customer segmentation, churn prediction, and inventory recommendations. The forecasting capability helps businesses estimate future demand and improve operational planning.

Through this project, practical knowledge of Python programming, machine learning, Streamlit dashboard development, and business analytics techniques was gained.

RetailPulse can be further enhanced with cloud integration and advanced AI models for real-time analytics and forecasting.

16. REFERENCES

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THANK YOU